

HOMEFINDER

Real Estate Assistance Web Application

COVER PAGE

Submitted to: Department of Computer Applications

Course: Bachelor of Computer Applications (BCA)

Project Title: HomeFinder - Real Estate Assistance Portal

Technology Stack: HTML5, CSS3, JavaScript, PHP 8.x, MySQL 8.x, Apache/XAMPP

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CERTIFICATE

This is to certify that the project titled "**HomeFinder - Real Estate Assistance Portal**" is a genuine work carried out by **Sakshi Shashikant Khopade**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA)**.

The project has been completed under the guidance of **Prof. Monali Meghal** and is submitted for evaluation.

Guide Signature: _____

Head of Department: _____

Date: _____

DECLARATION

I hereby declare that this project report titled "**HomeFinder - Real Estate Assistance Portal**" is my original work and has not been submitted previously in any form for the award of any degree, diploma, or other qualification.

Student Name: Sakshi Shashikant Khopade

Signature: _____

Date: _____

ACKNOWLEDGEMENT

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1. ABSTRACT

HomeFinder is a web-based real estate assistance portal designed to help users discover residential properties for rent or purchase in a structured and efficient way. The platform reduces the friction commonly experienced in property search by collecting user requirements in a standardized format, matching them against available listings, and enabling a controlled interest-confirmation flow for serious users.

The application supports three roles: guest visitor, registered client, and administrator. Guests can submit a quick requirement form and view a limited preview of matches. Registered clients can save detailed preferences, view full property details, mark interest in properties, and confirm a meeting through a simulated payment flow. Administrators can manage listings, view clients, track interests, and update meeting status.

This project demonstrates practical implementation of core web technologies and database-driven application design suitable for a BCA final-year submission.

2. INTRODUCTION

Property search in many cities is fragmented across classified portals, brokers, and informal channels. This often results in repeated conversations, irrelevant suggestions, and poor lead tracking. HomeFinder addresses that problem by:

- collecting requirements in a structured form,
- filtering listings by budget, area, BHK, and amenities,
- allowing serious users to confirm interest through a fee-based flow, and
- giving administrators a single place to manage leads and property data.

The application is designed to be simple enough for academic evaluation while still reflecting real-world system thinking.

3. PROBLEM STATEMENT

Traditional real-estate discovery is time-consuming and repetitive. A user usually has to:

- search multiple websites,
- contact brokers individually,
- repeat requirements many times, and
- follow up manually for site visits.

At the same time, property agencies struggle to track genuine leads, payment confirmations, and meeting schedules in a centralized way.

HomeFinder solves these issues by using:

- requirement-based search,
- session-based authentication,
- admin-controlled listings,
- an interest-confirmation flow, and
- centralized status tracking.

4. OBJECTIVES

The main objectives of the project are:

1. Provide a guest-friendly property search experience without mandatory login.
2. Allow registered clients to save detailed preferences and receive better matches.
3. Enable property discovery by area, transaction type, BHK, budget, and amenities.
4. Support interest confirmation through a simulated payment flow.
5. Give administrators control over property listings, client records, and meeting management.
6. Demonstrate secure and structured use of PHP, MySQL, HTML, CSS, and JavaScript.

5. SCOPE OF THE PROJECT

In Scope

- Landing page with hero section, statistics, how-it-works, and quick form
- Guest requirement submission and limited preview of matches
- Client registration and login
- Client dashboard with preference management
- Property listing and filtering
- Single property detail page
- Interest confirmation and simulated payment flow
- Payment success confirmation page
- Admin login and registration
- Admin dashboard
- Property CRUD
- Client management
- Interest and meeting tracking

Out of Scope

- Live payment gateway integration
- Mobile application
- Real-time chat
- Maps integration
- Automated email/SMS notifications
- Multi-language support
- Owner self-listing portal

6. USERS AND ROLES

Guest User

Guest users can explore the portal, submit a quick requirement form, and view a limited number of matched properties.

Registered Client

Registered clients can save preferences, access full property details, mark interest, and confirm a meeting through the booking flow.

Admin

Administrators manage the overall system, including listings, users, interest records, payment status, and meeting updates.

7. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum, 8 GB recommended
- Storage: 2 GB free space minimum
- Display: 1366x768 or higher

Software Requirements

- Windows or any Apache-compatible environment
- XAMPP with Apache and MySQL
- PHP 8.x
- MySQL 8.x
- Modern browser such as Chrome, Edge, or Firefox

8. TECHNOLOGY STACK

Front End

- **HTML5** for semantic structure
- **CSS3** for layout, responsiveness, and component styling
- **Vanilla JavaScript** for validation, counters, filtering, and AJAX interactions

Back End

- **PHP 8.x** for routing, authentication, validation, and database operations
- **MySQL 8.x** for persistent storage
- **Apache** via XAMPP as the web server

9. FUNCTIONAL FEATURES

Public Features

- Landing page with hero banner and animated counters
- Quick requirement submission form
- Featured properties section
- Testimonials
- Public property listings page

Client Features

- Registration and login
- Preference saving
- Full property browsing
- Property detail view
- Interest toggle
- Simulated payment and booking confirmation
- Dashboard with saved interests

Admin Features

- Admin login and protected registration
- Overview dashboard with statistics
- Property management with add, edit, delete
- Client list and client profile view
- Interest request list and meeting status updates

10. PAGE-WISE DESCRIPTION

10.1 Home Page

The home page introduces the platform, displays key metrics, and provides a quick-requirement form for guests.

Main sections:

- Navigation bar
- Hero section
- Statistics counters
- How-it-works section
- Quick requirement form
- Featured properties
- Testimonials

10.2 Client Registration Page

Users can create an account using name, email, phone number, and password. Passwords are securely hashed before storage.

10.3 Client Login Page

Registered clients authenticate using email and password. A session is started after successful login.

10.4 Client Dashboard

The dashboard displays saved preferences and matching properties. Users can also review their saved interests and meeting states.

10.5 Property Listing Page

The properties page provides filters for:

- transaction type,
- BHK,
- area,
- price range,

- amenities, and
- sorting options.

Guests see a limited preview, while logged-in users can browse more fully.

10.6 Property Detail Page

The detail page shows the property gallery, description, amenities, location, and a sticky interest call-to-action.

10.7 Payment and Confirmation Pages

The payment flow is simulated for project purposes. After confirmation, the system records an interest entry and payment record, then shows a success page with next steps.

10.8 Admin Pages

The admin section includes login, registration, dashboard, property management, client management, interest management, and status update functionality.

11. ARCHITECTURE OVERVIEW

HomeFinder follows a simple three-tier architecture:

1. **Presentation Layer**

Browser-rendered HTML, CSS, and JavaScript.

1. **Application Layer**

PHP pages and backend handlers process requests, validate data, and manage sessions.

1. **Data Layer**

MySQL stores users, admins, properties, preferences, search requests, interests, and payments.

Request Flow

1. The browser requests a PHP page.
2. PHP validates input and checks session state.
3. PDO executes prepared statements on MySQL.
4. The server returns HTML or JSON.
5. JavaScript enhances interaction where needed.

12. DATABASE DESIGN

12.1 users

Stores registered client accounts.

12.2 admins

Stores admin accounts separately for cleaner access control.

12.3 preferences

Stores saved client preferences in a one-to-one relationship with users.

12.4 properties

Stores all property listings, including type, BHK, price, area, furnishings, and amenities.

12.5 property_images

Stores multiple images for each property.

12.6 search_requests

Stores guest quick-form submissions.

12.7 interests

Stores confirmed interest requests and meeting status.

12.8 payments

Stores simulated payment records linked to interests.

13. ENTITY RELATIONSHIP SUMMARY

- One user has one preferences record.
- One user can create many search requests.
- One user can create many interests.
- One property can receive many interests.
- One interest has one payment record.
- One property can have many images.
- One admin can create many properties.

14. SECURITY AND VALIDATION

HomeFinder includes several security-oriented measures appropriate for a student project:

- Password hashing using `password_hash()`
- Password verification using `password_verify()`
- Session-based authentication
- Session ID regeneration after login
- Prepared statements through PDO
- CSRF token generation and verification
- Server-side validation for required fields and formats
- File upload validation for admin property images
- Restricted admin registration using a secret key
- Safe internal redirects

The simulated payment flow avoids exposing any real payment credentials while preserving the structure of a realistic booking workflow.

15. FRONT-END DESIGN APPROACH

The UI uses a clean, modern real-estate style with:

- semantic HTML5 markup,
- responsive CSS Grid and Flexbox layouts,
- card-based presentation for properties,
- sticky sidebars for filters and CTAs,
- animated counters on the home page, and
- consistent button, form, and badge styling.

The design prioritizes clarity, readability, and mobile responsiveness.

16. JAVASCRIPT FUNCTIONALITY

JavaScript is used for:

- form validation,
- animated counters,
- navigation toggles,
- property filter state handling,
- AJAX interest toggling,
- payment form interactions, and
- image gallery switching on the property detail page.

This keeps the interface interactive without requiring a heavyweight front-end framework.

17. KEY MODULES

`includes/init.php`

Initializes sessions, headers, timezone, and shared helpers.

`php/db.php`

Provides the PDO database connection and configuration constants.

`php/helpers.php`

Contains utility functions for redirects, sanitization, CSRF, formatting, and property matching.

`php/auth.php`

Handles client and admin login/registration flows.

`php/properties.php`

Processes property search and CRUD actions.

`php/interests.php`

Handles interest toggling and payment confirmation.

`php/requirements.php`

Processes guest quick-form submissions.

`php/preferences.php`

Saves client preferences.

18. TESTING SUMMARY

Manual Test Cases

1. Guest can submit the quick requirement form.
2. Guest preview results are limited to three properties.
3. Client registration stores hashed passwords and starts a session.
4. Client login redirects to the dashboard.
5. Preferences can be saved and reflected in matched results.
6. Property detail page shows protected address to logged-in users only.
7. Interest toggle respects paid interests and does not remove them.
8. Payment confirmation creates both interest and payment records.
9. Admin can add, edit, and delete properties.
10. Admin can filter interest records and update meeting status.

Validation Performed

- HTML forms use required fields and input constraints.
- JavaScript validation covers email, phone, password, and budget checks.
- PHP handlers enforce server-side validation and access control.

19. OBSERVATIONS

The project successfully demonstrates:

- structured user requirement collection,
- filtered property discovery,
- role-based access control,
- simulated booking/payment flow,
- admin-driven operations, and
- database-backed CRUD functionality.

The application is suitable for academic demonstration and can be extended into a production-grade portal with real payment integration, email notifications, and advanced search features.

20. FUTURE ENHANCEMENTS

Possible improvements include:

- live payment gateway integration,
- email and SMS notifications,
- map-based search,
- chat between client and agent,
- image upload previews,
- analytics charts for admin,
- pagination UX improvements,
- saved shortlist functionality, and
- PDF report generation for meeting confirmations.

21. CONCLUSION

HomeFinder is a complete real-estate assistance web application built using core web technologies and a relational database. It addresses a practical search and lead-management problem while remaining suitable for a third-year BCA final project.

The system demonstrates the ability to design a multi-role application, implement secure authentication, manage structured data, and build a responsive interface that is both functional and easy to understand.

22. APPENDIX

Project Structure

```
/homefinder/+- index.php+- login.php+- register.php+-  
dashboard.php+- properties.php+- property-detail.php+- payment.php+-  
- payment-success.php+- logout.php+- admin/+- php/+- includes/+-  
assets/+- sql/
```

Sample Property Locations

- Kothrud
- Baner
- Wakad
- Aundh
- Viman Nagar
- Hadapsar